SCU SEEDs Workshop

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This Talk

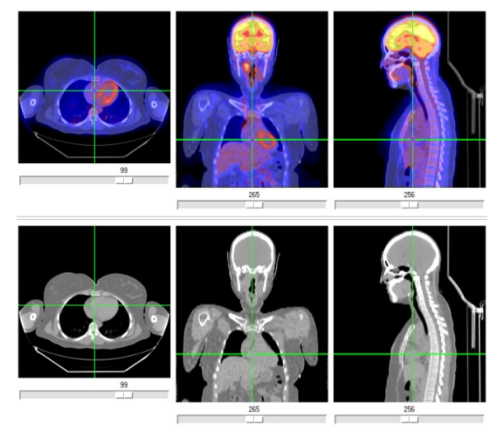
- Part I — Computing
- Part II — Computing at SCU
- Part III —Today’s activity

PART I — COMPUTING

What is Computing?

- Analysis, design and development of computer systems
- It is not just about programming
- It teaches you how to think more methodically and how to solve problems more effectively

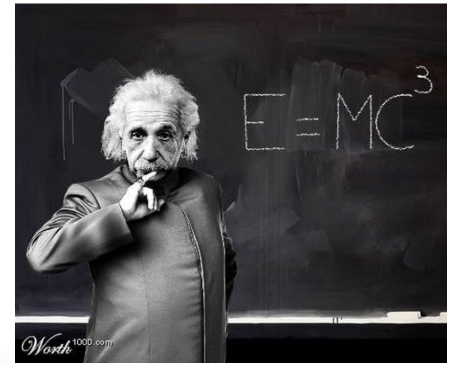
Computing is everywhere!



What is Computing?

- Computing includes a variety of fields:
  - Mathematics
  - Computer science
  - Computer engineering
  - Information science
  - Electrical engineering

What is Computing?



- What is a computer professional?
  - Will I have to grow fuzzy hair?

- What does s/he do?
  - Will I have to sit in front of a computer all day?

- What kind of people will I work with?
  - Will I have to become a geek nerd?

What is Computing?

- FUN, COOL, and EXCITING
  - Cutting edge projects
  - Exciting and talented people
  - All over the world, in every sector
  - Significant impact on society and our planet

Why Study Computing?

- **Intellectually interesting**
  - Logical reasoning and mathematical thinking
  - Possible workings of the human mind

Why Study Computing?

- **Computing supports and links to most other areas of study**
    - Computing and neuroscientists – the brain
    - Computing and Biologists – Genome
    - Computing and Meteorologists – weather prediction
  - Future scientists require basic knowledge of Computing

Why Study Computing?

- **Computing teaches problem solving**
  - Decomposition, abstraction, modular design
  - Analysis and design are carefully reviewed
  - Always new methods being investigated

Why Study Computing?

- **Computing builds teamwork and leadership skills**
  - Plan, organize, control, lead complex projects
  - Learn to deal with mix of talents
  - Estimate and deal with risk

Why Study Computing?

- **Computing develops life-long learning skills … *"Change is the only constant"***
  - Promotes learning to learn

"if GM had kept up with the technology like the computer industry has, we would all be driving $25.00 cars that got 1,000 miles to the gallon" – Bill Gates

  - Exponential growth makes many predictions look foolish

False Predictions

- “*I think there is a world market for maybe five computers*” -- Thomas J. Watson, founder and Chairman of IBM, 1943.

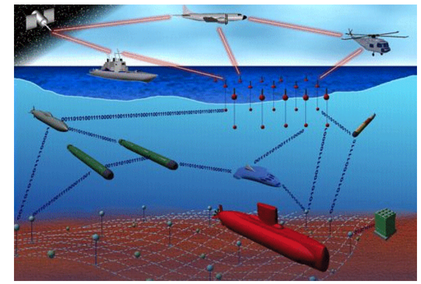
- “*Computers in the future may weigh no more than 1.5 tons*” -- *Popular Science*, 1949.

- “*640K ought to be enough for anybody*” -- Bill Gates, 1981.

Future Applications

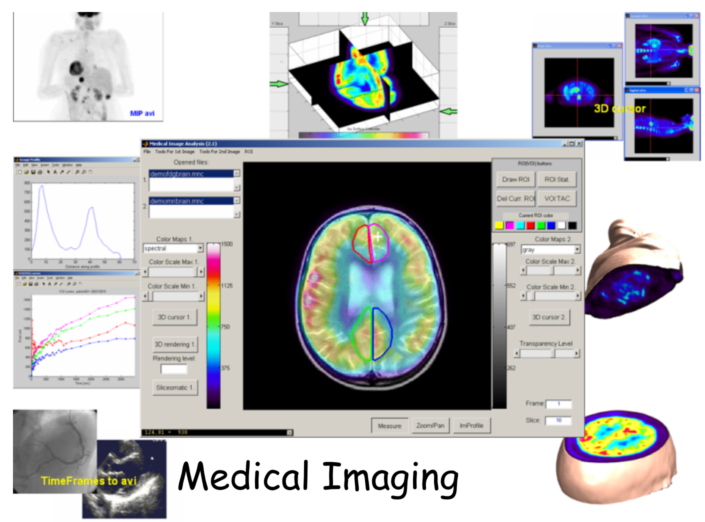
Self-driving car

Personalized Healthcare



Transforming the nation's defense

Internet of Things



Medical Imaging

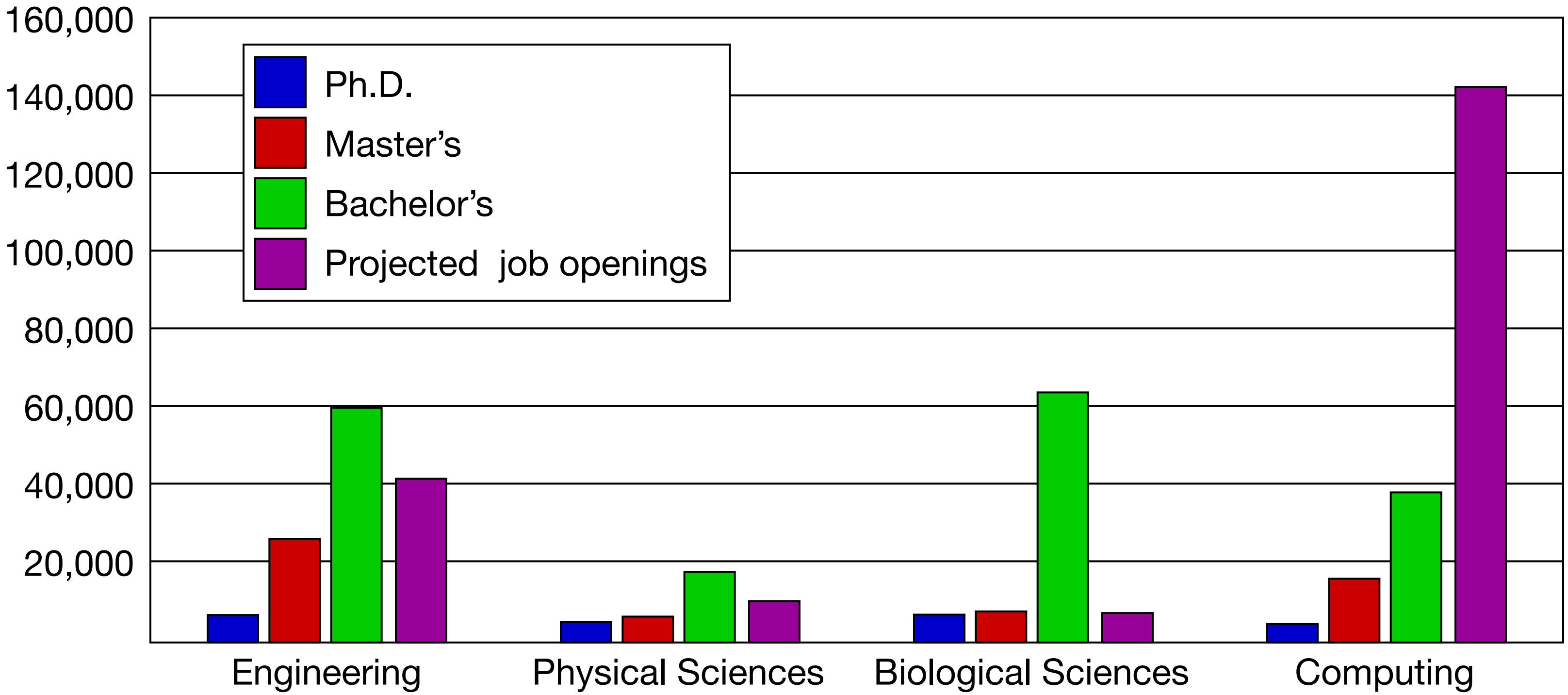
Comp. Science & Engineering

- Computer science
  - Often more mathematical
  - Computability theory
  - Algorithmic complexity
- Computer engineering
  - Often more hardware-oriented
  - Image and signal processing
  - Computer graphics

Career Opportunities

- System architect
- Network engineer
- Computer architect
- Software engineer
- Security specialist
- Game designer
- Test engineer
- Entrepreneur, musician, athlete, and more

Degree Production *vs.* Job Openings



Sources: Adapted from a presentation by John Sargent, Senior Policy Analyst, Department of Commerce, at the CRA Computing Research Summit, http://www.cra.org/govaffairs/content.php?cid=22.

Be Creative!

- Computing is the only tech field in which you can create a product from scratch and commercialize it independently

- Computing is the only tech field in which you can easily come up with your own personal solutions.

PART II — COMPUTING AT SCU

Computing Degrees at SCU

- Undergraduate degrees
  - Computer science and engineering (CSE)
  - Web design and engineering (WDE)
  - Mathematics and computer science
- Graduate degrees
  - Computer science and engineering
  - Software engineering
- 5-year Master’s program

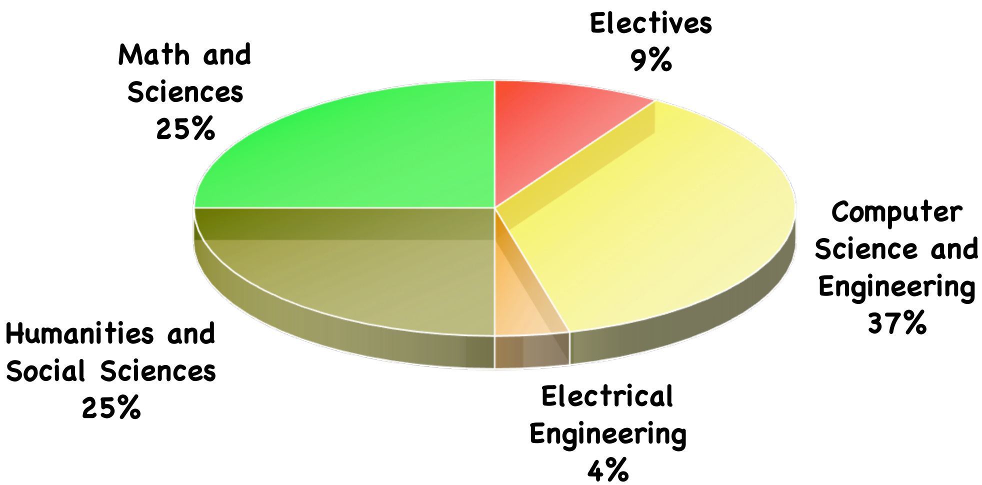
Undergraduate CSE

- Combination of computer science and computer engineering
- Focuses on theoretical and practical aspects of computing
- Design and construction of both hardware and software systems
  - Computer networks, operating systems, algorithms, compilers, software engineering, embedded programming, Web programming, robotics, 3D animation

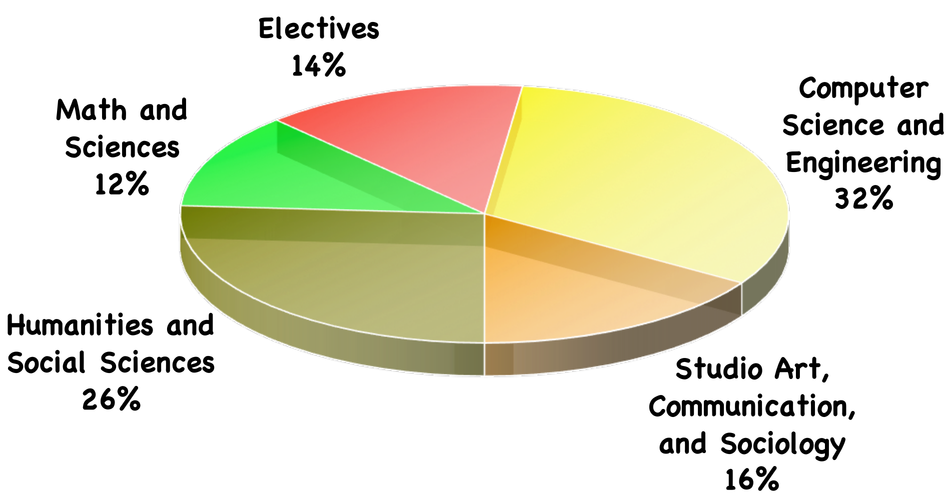
Undergraduate WDE

- New major started in 2009
  - One of the first such programs in the country
- Combines computing with other disciplines:
  - Graphic arts
  - Communication
  - Sociology
- What will these specialized graduates do?
  - Improve Web infrastructure
  - Develop interactive, multimedia content
  - Analyze the huge amount of information on the Web (Big data)
  - Understand the societal impact of the Web

Coursework



Computer Science and Engineering



Web Design and Engineering

Where Will You Work?

- Recent graduates went to work for:
  - Cisco, Apple, Microsoft, IBM, Google, Facebook, Groupon, Amazon, Anritsu, F5 Networks
  - Starting salary range: $70K–$100K
- Recent graduates also continued their education:
  - Ph.D program at Berkeley, UCSD, etc.
  - M.S. programs at SCU, CMU, Stanford, etc.